Water resources are potentially useful to humans. Water in India is available in three sources. They are **precipitation, surface water** and **groundwater**.

**Table 1: Estimated Volume of Water on the Earth's Surface**

| Water Source | Volume of water (Cubic Miles) | Percentage to Total Water |
|---|---|---|
| Oceans, Seas, & Bays | 321,000,000 | 96.54 |
| Ice caps, Glaciers, & Permanent Snow | 5,773,000 | 1.74 |
| Groundwater | 5,614,000 | 1.69 |
| Soil Moisture | 3,959 | 0.001 |
| Ground Ice & Permafrost | 71,970 | 0.022 |
| Lakes | 42,320 | 0.013 |
| Atmosphere | 3,095 | 0.001 |
| Swamp Water | 2,752 | 0.0008 |
| Rivers | 509 | 0.0002 |
| Biological Water | 269 | 0.0001 |

(*Source: Shiklomanov, 1993*)

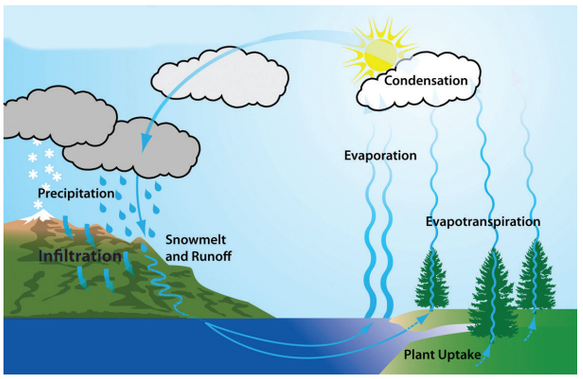
## Hydrological Cycle or Water Cycle

Hydrological is the science which deals with the various aspects of water such as its occurrence, distribution, movement and properties on the planet earth. Availability of water on the earth is not uniform. Some places are very rich and some places are poor in water resources.

Hydrological cycle is a global sun-driven process where water is transported from oceans to atmosphere, from atmosphere to land and from land back to oceans. The water cycle can be considered as a closed system for the earth, as the quantity of water involved in the cycle is invariable, though its distribution varies over space and time.

Evaporation takes place from the surface water and transpiration from the plants. Water vapour gets condensed at higher altitudes by condensation nuclei and form clouds. The clouds melt and sometimes burst resulting in precipitation of different forms. A part of water from precipitation flows over the land is called runoff and the other part infiltrates into the soil which builds up the groundwater.

Hydrological cycle is a circulation of water. It is a continuous process and takes place naturally. The three important phases of the hydrological cycle are: **1) Evapotranspiration, 2) Precipitation** and **3) Runoff**.




**Hydrological Cycle**

## Components of Hydrological Cycle

There are six main components in hydrological cycle. They are: 1) Evapotranspiration, 2) Condensation, 3) Precipitation, 4) Infiltration, 5) Percolation, and 6) Runoff.

### Evapotranspiration

It is defined as the total loss of water from the earth through evaporation from the surface water bodies and the transpiration from vegetation. In cropped area, it is difficult to determine the evaporation and transpiration separately. Therefore it is collectively called as evapotranspiration. The following part explains the process of evaporation and transpiration separately.



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